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PALAEOMAGNETISM OF CRETACEOUS AND JURASSIC VOLCANIC ROCKS IN WEST SICILY

A. SCHULT

Institut für Allgemeine und Angewandte Geophysik der Ludwig-Maximilians-Universität München, Munich (Federal Republic of Germany)

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The mean palaeomagnetic pole position obtained from Upper Cretaceous rocks in west Sicily is at 21°N, 100°E (A_{95} = 15°), and at 38°N, 67°E (A_{95} = 31°) obtained from Middle Jurassic rocks. These pole positions are completely different from comparable pole positions for southeast Sicily and Africa and imply a clockwise rotation of west Sicily since the Upper Cretaceous of about 90° relative to southeast Sicily and Africa and also a clockwise rotation of about 60° relative to "stable" Europe. The sense of rotation of west Sicily is opposite to any known rotation of other crustal blocks in the central Mediterranean.

1. Introduction

Previous palaeomagnetic investigations have been carried out in southeast Sicily at Capo Passero on Upper Cretaceous volcanic rocks by Schult [1], Barberi et al. [2] and Gregor et al. [3], and also on Upper Tertiary volcanic rocks in east Sicily [2,3]. In this paper palaeomagnetic results for Upper Cretaceous and Middle Jurassic volcanic rocks in west Sicily are reported.

It is of interest to compare palaeomagnetic data from west Sicily with those from southeast Sicily because it was suggested by various authors that ancient plate boundaries may cross Sicily (e.g. [2,4,5]). In Fig. 1 probable boundaries according to Caire [5] are shown. The main structural features of west and north Sicily are several nappe units (e.g. [5,6]) showing geological allochthony whereas southeast Sicily (the so-called Ragusa platform) is relative stable and appears to a prolongation of the Sahara platform.

2. Sampling and measurements

There are several outcrops of Mesozoic volcanic rocks in west Sicily. A compilation is given by Floridia [7]. With the exception of the largest basalt flow at Guiana (Fig. 1, site 14) most outcrops are very small and often not larger than a few square meters.

The outcrops are described as follows (see Fig. 1). Site 1: immediately southwest of Custonaci at the road to Trapani. Site 2–5: about 2 km southwest of Custonaci along the river Torrente Forgia. Site 1 is a pillow basalt, site 2–4 are dykes, site 5 is basaltic

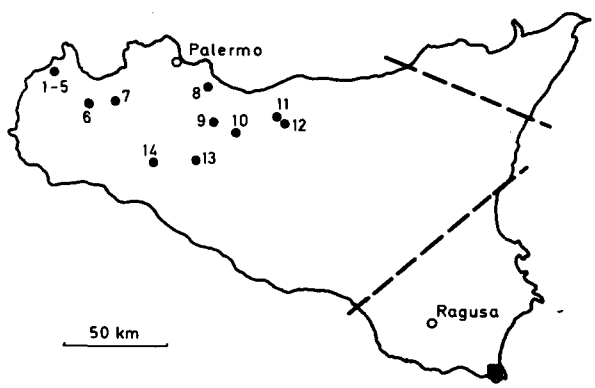


Fig. 1. Solid circles represent the sampling sites of Upper Cretaceous (sites 1–5) and Middle Jurassic volcanic rocks in west Sicily. Solid square represents the Upper Cretaceous volcanic rocks in southeast Sicily at Capo Passero. Thick broken lines indicate possible boundaries between crustal blocks according to Caire [5].

breccia inbedded in tuff. A detailed petrographic and mineralogical description is given by Vianelli [8]. Site 6: east side of Monte Barbero (Segesta) in the grotto. The basaltic rock is strongly weathered [9]. Site 7: at the east side of Monte Bonifato (Acamo) in the large quarry. This basalt is also strongly weathered [9]. Site 8: dyke, 3 km south of Casteldaccia, east side of Rocca di Papa. Site 9: basalt flow ca. 500 m north of Vicari. Site 10: basalt flow, 1 km north of the railway station of Roccapalumba. Site 11: dyke, 1 km northeast from Scillato at km 23.3 of the road Collesano-Polizzi. Site 12: dyke 3.5 east-southeast from Scillato 500, north of km 27.5 of the road Collesano-Polizzi in a gorge. Site 13: basalt intrusion at the southern end of the dam of Lago Piano del Leone. Site 14: basalt flow at the street ascending to Guiana from street Nr. 188.

According to Vianelli [8], the volcanic rocks near Custonaci (sites 1–5) are of Upper Cretaceous age. The basaltic rocks near Vicari (site 9) and Roccapalumba (site 10) are of Bajocian age (Middle Jurassic) [7,10,11]. The other basaltic rocks are not so well dated but are probably also of Middle Jurassic age [11].

Oriented cores (2.5 cm diameter) were taken by means of a portable core drilling machine. The remanent magnetization was measured with a fluxgate spinner magnetometer. Pilot specimens from each site were chosen for stepwise demagnetization in alternating magnetic fields up to 500 Oe. All remaining samples were then treated in appropriate alternating fields according to the results of the pilot specimens. The cleaned remanence directions of samples from each site were averaged to obtain site mean directions. In some cases a tilt correction was necessary, estimated from the adjacent strata. The results are shown in Table 1, the directions of magnetization with their respective circles of confidence are shown are shown in Fig. 2.

Only one Jurassic site had reversed remanence

TABLE 1

Site mean palaeomagnetic results after AF cleaning (location is at 38°N, 13°E)

Site	N	AF (Oe)	D (degrees)	I (degrees)	α_{95} (degrees)	k	°N	°E
1	8	220	69.4	+30.8	4.5	152	26.1	102
2	12	200	64.6	+17.2	4.6	89	25.3	112
3	9	130	73.7	+42.2	9.4	31	27.1	92
4	4	350	91.6	+23.2	10.6	76	6.1	92
5	21	unstable						
6	19	unstable						
7	14	220	254.6	–40.6	15.9	7	25.7	93
8	6	unstable						
9	10	250	101.2	+62.3	16.5	10	18.2	61
10	14	250	269.4	+84.1	10.3	16	36.9	358
11 *	9	120	296.7	+19.0	13.5	16	27.0	274
12	8	180	109.5	+70.5	10.0	31	20.3	48
13	5	150	53.7	+78.7	19.7	16	48.0	40
14	16	120	46.7	+36.7	3.5	106	46.1	113
Upper Cretaceous (sites 1–5)								
Mean:	4		74.8	+28.7	17.4 (A_{95} = 14.9°, dp = 10.6°, dm = 19.1°)	29	21.2	100
Middle Jurassic (sites 6–14)								
Mean:	6		72.7	+66.0	21.6 (A_{95} = 31.5°, dp = 23.1°, dm = 39.8°)	10.5	37.9	67

N = number of samples, AF = peak value of alternating field cleaning (10^4 Oe = 1 T), D ≙ declination, I = inclination, α_{95}, A_{95} = radius of the 95% circle of confidence, k = precision parameter.
* Eliminated site.

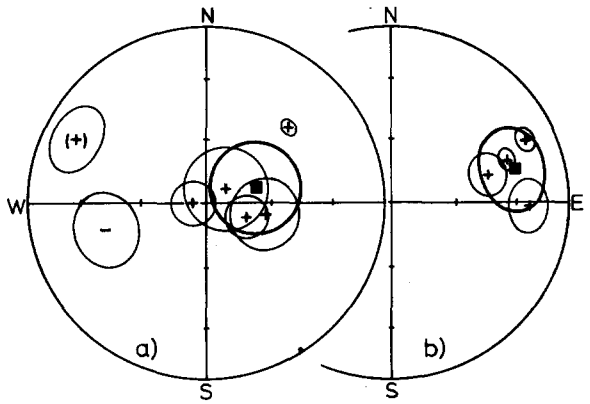


Fig. 2. Stereographic projection of mean site directions (+, −) and overall mean (■) with respective circles of 95% confidence: (a) Middle Jurassic rocks, (b) Upper Cretaceous rocks. The site in brackets was discarded because of intermediate direction.

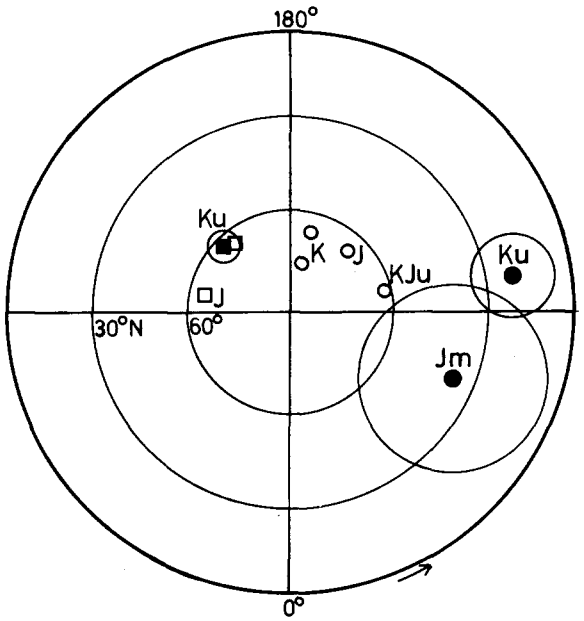


Fig. 3. Cretaceous and Jurassic palaeomagnetic pole positions with their respective circles of confidence (when given). K: Cretaceous, J: Jurassic, u: Upper, m: Middle. ● = west Sicily (this paper); ■ = southeast Sicily [1,2]; ○ = "stable" Europe: K [14,15], KJu [16], J [18]; □ = Africa: Ku [17], J [18].

direction all others had normal directions. The magnetization of three sites was unstable and several sites have relatively large within-sites scatter. The initial intensities ranged between 10 and 1000 × 10^{-5} gauss (10^{-3} G ≙ 1 A m^{-1}) and were reduced up to an order of magnitude after alternating field (AF) cleaning. Giving unit weight to each site direction, the mean directions and palaeopole positions for the Upper Cretaceous and Middle Jurassic rocks were calculated (see Table 1). Site 11 was discarded from the mean because of intermediate direction.

3. Discussion

Due to the restricted number of suitable volcanic rocks and possibly due to other reasons (e.g., uncertain tilt correction, relative unstable remanence) the circles of confidence for the means are relatively large, in particular for the Jurassic rocks. But it is assumed that the data are reliable enough to provide some useful information.

In Fig. 3 Cretaceous and Jurassic pole positions for west Sicily, southeast Sicily, Africa, and "stable" Europe are compiled. The Upper Cretaceous pole positions for west and southeast Sicily are completely different from each other. This is consistent with the suggestion that old plate boundaries cross Sicily and that southeast Sicily and west Sicily previously belonged to different crustal blocks [2,4,5]. Comparison of the mean direction of magnetization of Cretaceous rocks from west Sicily ($D = 75°$, $I = 28.7°$) with that for southeast Sicily ($D = 347°$, $I = 23°$) implies a clockwise rotation of west Sicily of about 90° relative to southeast Sicily since the Upper Cretaceous because a rotation of 90° would bring the Upper Cretaceous pole for west Sicily to 63.7°N, 227°E into coincidence with the palaeopole for southeast Sicily at 62.5°, 224°E. The same 90° angle clockwise rotation of west Sicily since the Upper Cretaceous is also found relative to Africa. A similar clockwise rotation of 70° with respect to Africa and southeast Sicily since the Mesozoic was proposed by Caire [5] in a hypothetical reconstruction of the central Mediterranean region. The coincidence of the Upper Cretaceous positions for southeast Sicily and Africa is consistent with the suggestion that southeast Sicily (the Ragusa platform) is a prolongation of the Sahara platform (e.g. [5,6]).

The rotation of west Sicily since the Upper Cretaceous relative to "stable" Europe is clockwise (see Fig. 3) and is of the order of 60°. This figure is only tentative because Cretaceous pole positions for "stable" Europe are not very reliable. Similar amounts of clockwise rotation of west Sicily relative to Africa (and therefore also with respect to southeast Sicily) and also relative to "stable" Europe indicate the Jurassic pole positions for west Sicily, Africa and Europe (see Fig. 3). Reliable amounts of rotation cannot be given because the circle of confidence for the west Sicily pole position is very large. It is known that according to palaeomagnetic data most crustal blocks in the central Mediterranean underwent counterclockwise rotation which was recently also found for northern Calabria [12]. Counterclockwise rotation implies sinistral shear. Illies [13] has pointed out that temporarily also dextral shear occurred in the Mediterranean which may have caused the clockwise rotation of west Sicily.

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